WorksOnMyMachine

Mind Reading Using Machine Learning and EEG



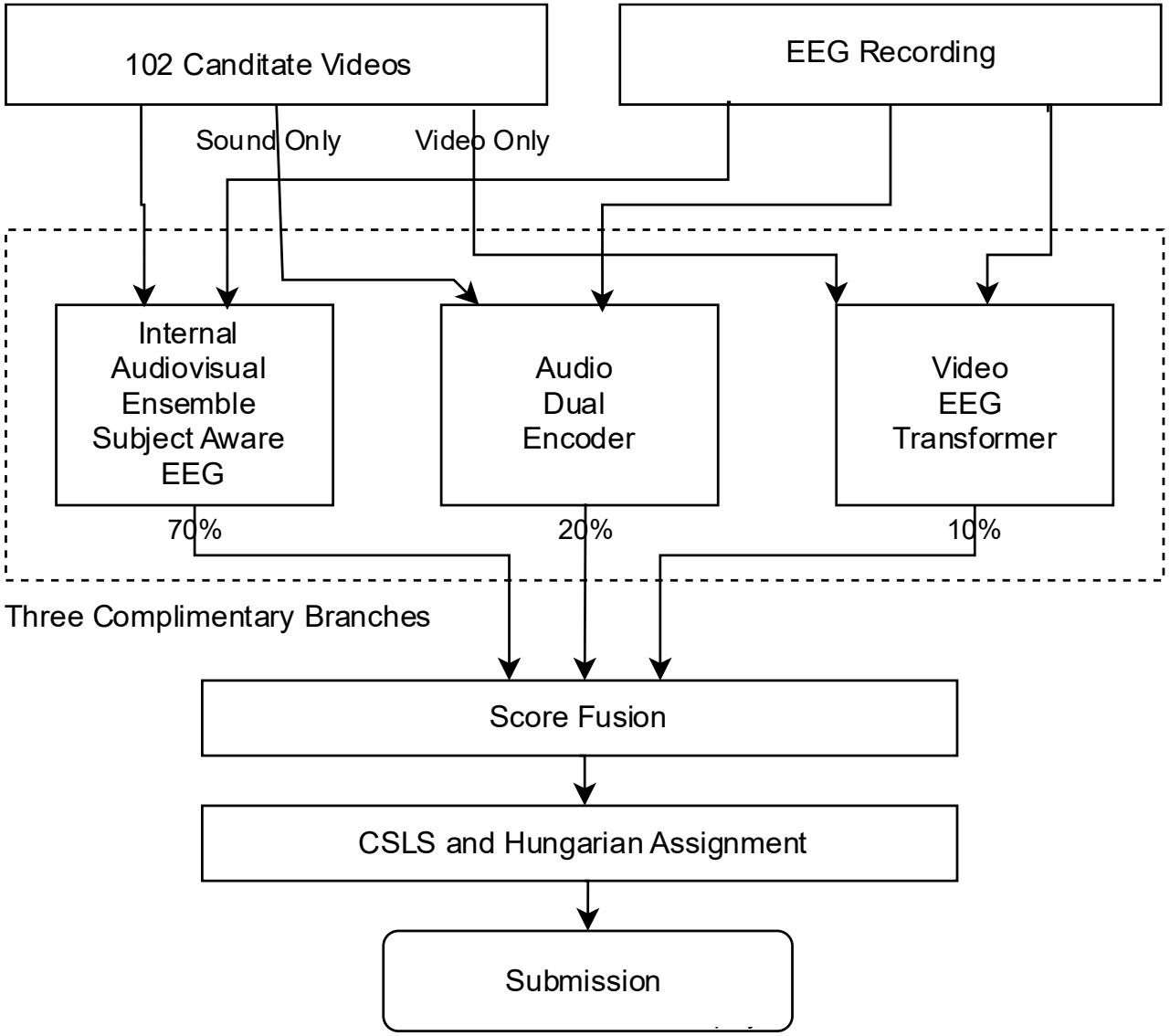
Our Approach

- ***CORTIVA**: CANDIDATE-SCORE FUSION*[1] as base architecture

- Filter EEG and adapt to each participant
- Preserve timing in audio and video features
- Train EEG and stimulus encoders to match
- Combine model scores and assign videos

[1] J. Wang and K. Chen, “CORTIVA: Candidate-Score Fusion of Complementary Visual Teachers for EEG- and MEG-to-Image Retrieval,” arXiv preprint arXiv:2608.01355, 2026. doi: 10.48550/arXiv.2608.01355.

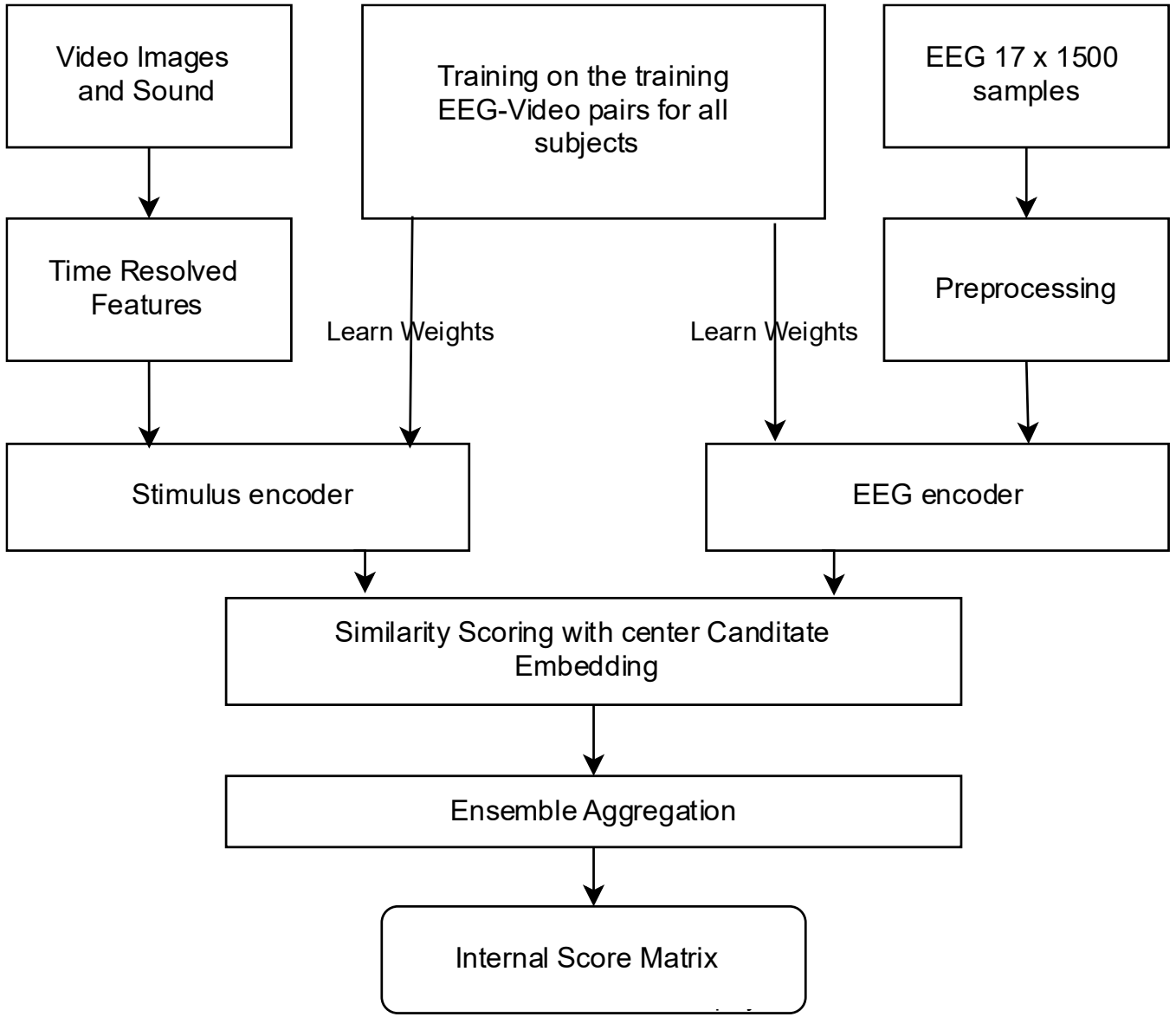
Inference - Simplified



Three seperate complimentary Models

- Self Trained Video EEG Transformer (with THINGS-EEG2 pretraining)
- Self Trained Audio EEG Transformer
- One To One Assignment (Only top1) and CSLS
  => Assigned video first; remaining in CSLS order.
- OpenAi Whisper as Additional Audio Encoder

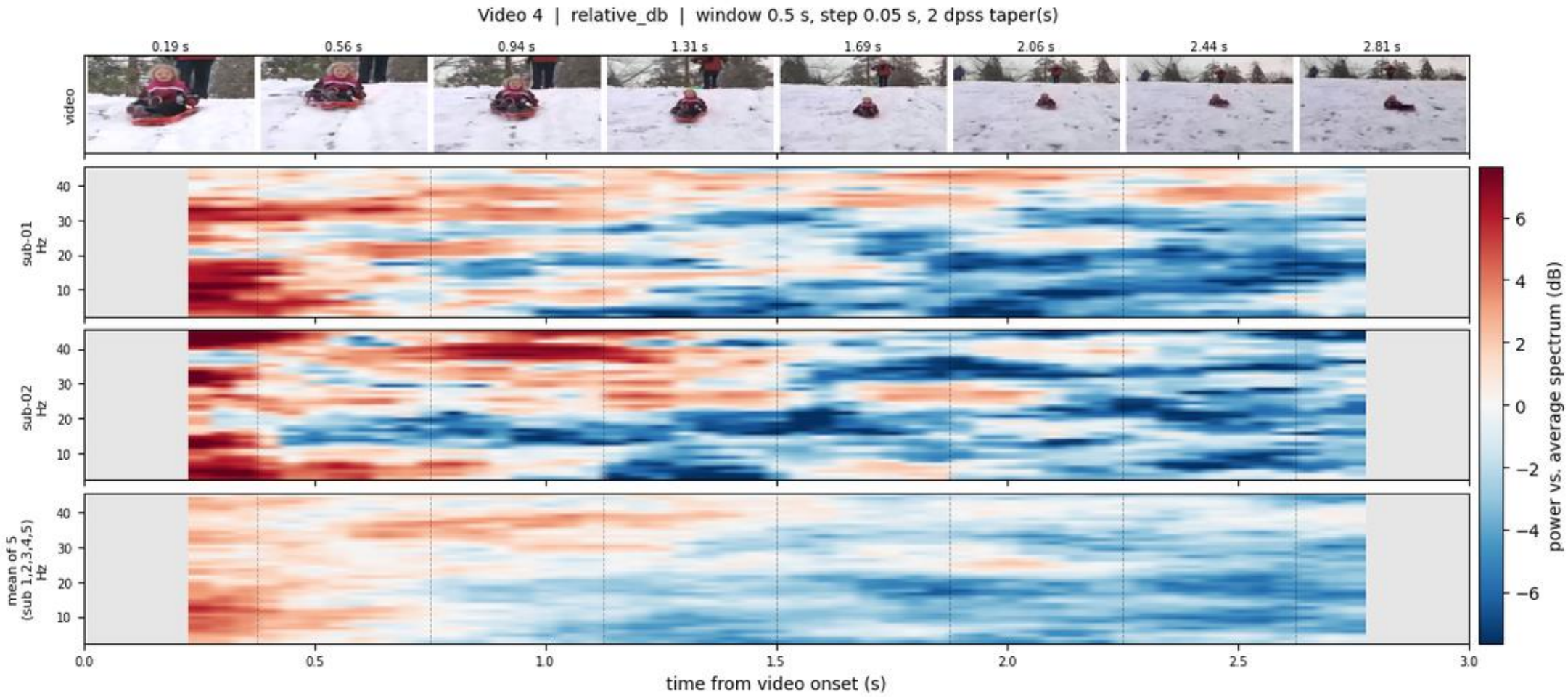
Internal Audio Visual Ensemble



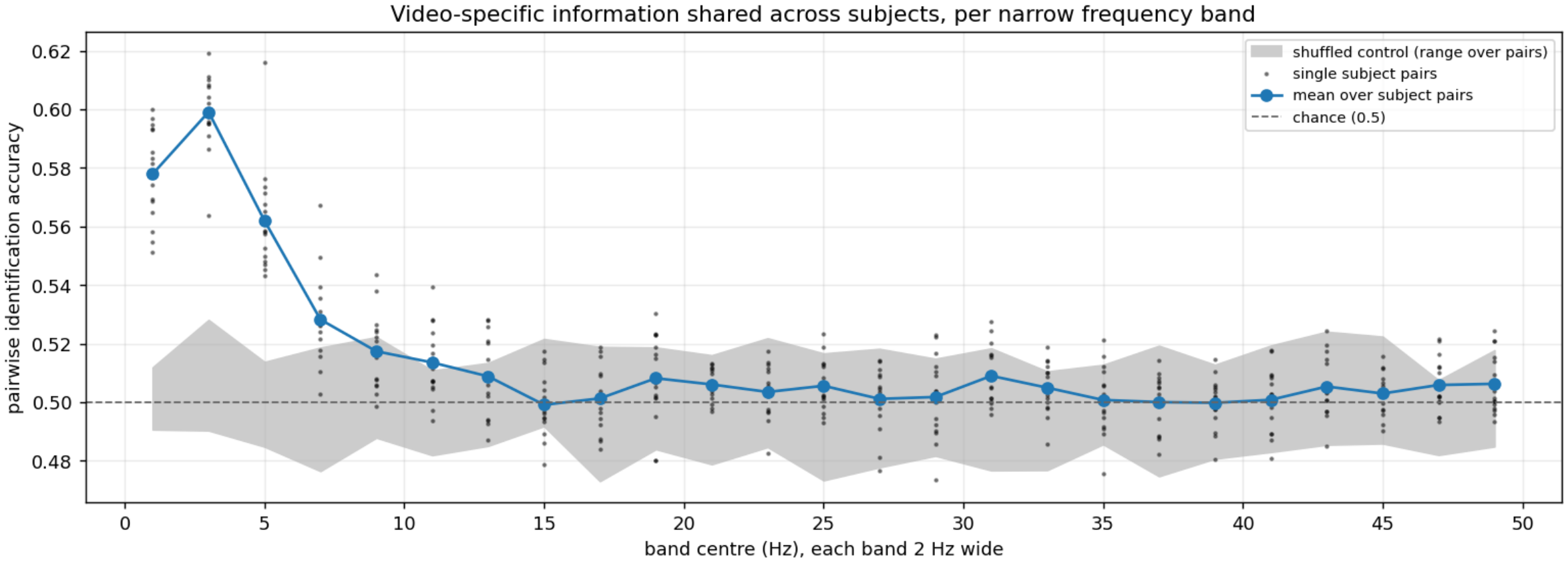
Internal Model that accounts for 70 percent of the final score

- Encodes Video data and EEG data into intermediate Representations
- Compares the IRs
- Scores with Ensemble of Models that are Subject-Aware (Candidate Embedding)

Data Analysis



Data Analysis

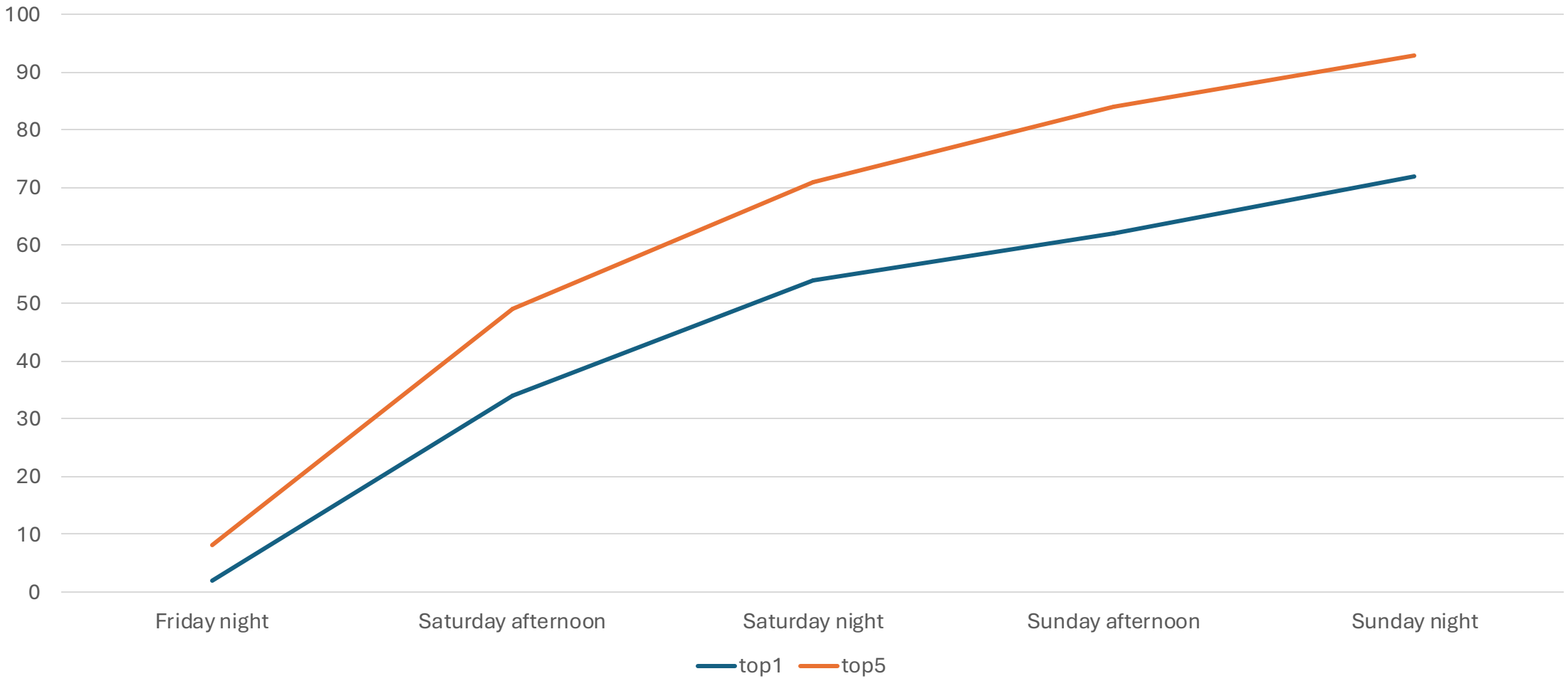


Final Results

Evaluation	Top-1	Top-5	MRR
Final v7 DEV: 408 queries, 102 candidates	72.55%	93.87%	0.8196
Training-video validation: independent ranking	29.88%	54.11%	0.4173
Training-video validation: CSLS and assignment	29.15%	58.25%	0.4257

18x improvement over Baseline of 3,9%

Our Results Over Time



Acknowledgements

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